LEC26 - The Integral Test

Thursday, March 13, 2025 Section 5.3

Warm-up Problem

Does the series $\sum_{n=1}^{\infty} \frac{n^2}{4n^2 - n}$ converge or diverge

⇒ Taking the limit as $n \to \infty$ gives $\lim_{n\to\infty}\left[\frac{n^2}{4n^2-n}\right] = \frac{1}{4} \neq 0$

⇒ Therefore, since $\lim_{n\to\infty} a_n \neq 0$, the series must diverge

Example

Does the series $\sum_{n=1}^{\infty}\left[\frac{4}{n^3} + \frac{1}{n^{1.1}}\right]$ converge or diverge? Why?

⇒ This series is a sum of p series where $p_1 = 3 > 1$, $p_2 = 1.1 > 1$

⇒ Therefore, since both p-series have $p > 1$, both p-series converge

⇒ Therefore, the series must converge

Theorem 5.9 - The Integral Test

Suppose $\sum_{n=1}^{\infty} a_n$ is a series with positive terms

Suppose f is a function such that $f(n) = a_n$ for all $n \geq N$ (for some positive integer N). If f is positive, continuous, decreasing for $x \geq N$, then $\sum_{n=1}^{\infty} a_n$ and $\int_N^{\infty} f(x)\,dx$ either both converge or both diverge

Example

Does the series $\sum_{n=1}^{\infty}\left[\frac{8}{3+4n}\right]$ converge or diverge?

⇒ Let $f(x) = \frac{8}{3+4x}$, $x \geq 1$

⇒ We know that $f(x)$ is positive, continuous (for $x \geq 1$) and decreasing

⇒ Consider $\int_1^{\infty} f(x)\,dx = \lim_{t\to\infty}\left[\int_1^t \frac{8}{3+4x}\right]$

→ let $u = 4x$, so $\frac{du}{4} = dx$

→ $\lim_{t\to\infty}\left[\int_4^{4t} \frac{8}{3+u}\cdot\frac{1}{4}du\right] = \lim_{t\to\infty}\left[2\int_4^{4t}\frac{1}{3+u}du\right] = \lim_{t\to\infty}\left[2\left[\ln|3+u|\right]_4^{4t}\right]$

$= \lim_{t\to\infty}\left[2\ln|3+4t| - 2\ln|7|\right] = \infty$

⇒ Therefore, since $\int_1^{\infty} f(x)\,dx$ diverges, then $\sum_{n=1}^{\infty}\frac{8}{3+4n}$ must also diverge by the integral test

Example

Is the following solution correct? If not, find all errors (there may be multiple)

Problem: Does the series $\sum_{n=0}^{\infty}\frac{1}{n^2+1}$ converge or diverge?

1) Let $f(x) = \frac{1}{x^2+1}$ ✓

2) $f'(x) = \frac{-2x}{(x^2+1)^2} < 0$ for $x > 0$. Since $f'(x)$ is negative, f is decreasing for $x \geq 0$ ✓

3) Lets use the integral test note: didn't verify that f is positive (it is, so it will technically still work)

4) $\int_0^{\infty}\frac{1}{x^2+1}dx = \lim_{t\to\infty}\int_0^t \frac{1}{x^2+1}dx = \lim_{t\to\infty}\left[\arctan(x)\right]_0^t$ ✓

5) $\lim_{t\to\infty}\left[\arctan(t) - \arctan(0)\right] = \frac{\pi}{4} - 0 = \frac{\pi}{4}$ $\lim_{t\to\infty}\left[\arctan(t)\right] = \frac{\pi}{2}$

6) So the integral converges to $\frac{\pi}{4}$

7) Therefore, the integral test tells us the series also converges to $\frac{\pi}{4}$ The integral test says nothing about what the series converges to, only whether the series converges